

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (ME) Examination

April-May 2018

ME 307 Refrigeration & Air-conditioning

Date: 03.05.2018, Thursday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator, steam table and psychrometric chart is allowed.

SECTION – I

Q - 1 Answer the questions below.

- (i) At what state on psychrometric chart, Dry bulb, Wet bulb and Dew point temperatures are identical? [01]
- (ii) Can water vapour in the air be treated as ideal gas? Give your opinion. [02]
- (iii) How will specific humidity and relative humidity of air changes in the well shield room, when it is heated? [02]
- (iv) How does the air motion in the vicinity of human body affect the human comfort in extreme summer? [02]

Q-2.a Define specific humidity. Derive psychrometric relation of it in terms of vapour pressure. [04]

Q – 2.b Answer any two questions. [10]

- (i) What is Wet bulb temperature? How wet bulb temperature of atmosphere is measured? Why it is different from dry bulb temperature? Draw the necessary diagram.
- (ii) Draw the line diagram of All water central air conditioning system and compare its performance with all air system.
- (iii) State your opinion on the factors and its effects on the human comfort.

Q - 3 Answer any two questions. [14]

- (i) Room air at 20 °C DBT and 50% RH is mixed with outdoor air at 40 °C DBT and 30% RH in the ratio of 4:1. The mixture is passed through a cooling coil whose surface temperature is 10 °C and by-pass factor 0.2. Calculate: (a) Condition of air before entering the cooling coil (b) Condition of air leaving the coil (c) If 250 m³/min of air is supplied to the room find the refrigeration load on cooling coil.
- (ii) An air conditioned plant is designed for a small office for winter condition: Outdoor air condition = 10 °C DBT and 8 °C WBT; Indoor air condition = 20 °C and

60% RH; Amount of free air circulated = $15 \text{ m}^3/\text{min}$. The indoor condition is achieved first by sensible heating and then adiabatic humidification at constant WBT. Calculate:

- (a) Capacity of Heater in kW and its surface temperature if the by-pass factor is 0.32.
 (b) Capacity of humidifier and humidifier efficiency.
- (iii) $100 \text{ m}^3/\text{min}$ of air at 15°C and 80% RH is heated until its temperature becomes 22°C without changing its moisture content. Calculate: (a) mass of air circulated per min (b) Heat added to air per minute (c) specific humidity of entering air (d) vapour pressure of entering air (e) RH of heated air. Barometer Reading = 740 mm of Hg. **Solve the problem without using psychrometric chart.**

Temperature ($^\circ\text{C}$)	Saturation pressure (bar)	Enthalpy of evaporation
11	0.0136	2475.5 kJ/kg
15	0.017	2466.1 kJ/kg
22	0.0264	2449.6 kJ/kg

SECTION – II

Q - 4 Answer the questions below.

- (i) Does the VCR cycle involve any internal irreversibility? Give your opinion. [02]
 (ii) Why is throttling valve not replaced by isentropic turbine in the ideal VCR cycle? [02]
 (iii) Why ammonia is not used as a refrigerant in domestic refrigerator even though it is having good thermal properties? [01]
 (iv) Give chemical formula for R22 and R756. [02]

Q – 5.a A dense close air refrigeration system working between 4 bar and 16 bar extract 126 MJ of heat per hour. Air enters into compressor at 5°C and into expander at 20°C . Assuming unit runs at 300 rpm. Find out: (a) power required to run the unit and (b) bore of compressor. Assumption: (1) Compressor and expander are double acting and stroke length is 300 mm. (2) Mechanical efficiency of compressor is 80% and that of expander is 85%. (3) Compression and expansion processes are isentropic. Consider bell-coleman cycle. [05]

OR

- Q – 5.a** Draw p-V and T-s diagram of bell-colleman cycle and derive C.O.P. formula for the same. [05]
Q – 5.b Discuss thermodynamic properties of ideal refrigerant. [04]
Q – 5.c Derive the equation of C.O.P. for ideal VAR cycle with block diagram. [05]

OR

Q – 5.c Draw the block diagram of LiBr-H₂O VAR system and explain its working. [05]

Q – 6.a Explain actual vapour compression refrigeration cycle with T-s and p-h diagram. [07]

Q – 6.b A Refrigeration machine using R-12 as refrigerant operates between the pressures 2.5 bar and 9 bar. The compression is isentropic and there is no sub-cooling in condenser. The vapour is dry and saturated at the beginning of compression. If the actual C.O.P. is 0.65 times theoretical C.O.P. Calculate the net cooling produced per hour. The refrigerant flow is 5 kg/min. Take C_p for superheated vapour is 0.64 kJ/kg K. Draw p-h and T-s diagram of the cycle. [07]

Pressure (bar)	Saturation Temperature (°C)	Enthalpy (kJ/kg)		Entropy of saturated vapour (kJ/kg K)
		h_f	h_g	
9	36	70.55	201.8	0.6836
2.5	-7	29.62	184.5	0.7001

OR

Q – 6.a Sketch and explain a cascade refrigeration system with pressure enthalpy diagram. [07]

Q – 6.b Explain the effect of (a) suction pressure and discharge pressure (b) superheating and sub-cooling on the C.O.P. of VCR cycle. [07]
